

SMART INDIA HACKATHON 2026 — TITLE PAGE

- **PROBLEM STATEMENT ID:** SIH2026_CYBER_014
- **PROBLEM STATEMENT TITLE:** Real-Time AI Voice Clone Detection Platform
- **THEME:** Cyber Security & AI Safety
- **PS CATEGORY:** Software
- **TEAM ID:** SIH2026_TEAM_DHVANI
- **TEAM NAME:** Dhvani Rakshak



IDEA TITLE: Dhvani Rakshak — Solution & Extra Features

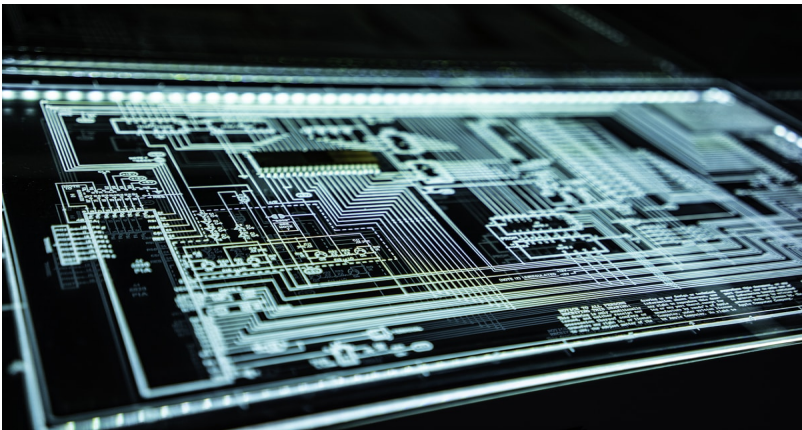
❖ Core Features vs 4 Extra Innovations Added

■ CORE SYSTEM FEATURES:

- Real-Time Call Scanner (< 200 ms latency)
- AI Scam Attack Call Simulator Sandbox
- Executive Security Dashboard (■2.45 Cr Saved)
- 10-Sec Privacy Enrollment Studio (Zero Audio Saved)
- Auto-Block Policy Configurator
- DPDP Act 2023 & RBI Safety Audit Logs

■ 4 EXTRA ADDED FEATURES:

- ■ Blockchain Voice Pass (Polygon SHA-256 court proof)
- ■ Behavioral Biometrics (Keystroke, mouse & stress dynamics)
- ■ Govt VIP Protection (IAS/IPS emergency police webhooks)
- ■ Explainable AI (XAI) ('Flat pitch' plain-English cards)



TECHNICAL APPROACH — 4-Step Architecture Flowchart

[Step 1: Audio Ingestion]: Live Phone Call WebSockets tap -> 16kHz mono normalization.

[Step 2: Feature Extraction]: 500ms Sliding Window -> Extract 40 MFCCs & pitch jitter.

[Step 3: AI Classification]: Deep Neural Model computes Risk Score (0 - 100%).

[Step 4: Action & Audit]: Auto Call Cut / Step-up OTP -> Polygon Blockchain Merkle proof.



FEASIBILITY AND RISK MITIGATION GRID

■ **Sub-200ms Speed:** Real-time call stream interception

■ **Zero Storage Overhead:** RAM volatile processing only

■ **Extra XAI Clarity:** Plain-English fraud explanation cards

• Lossy 2G/4G Cellular Compression --> Strategy: 16kHz F0 pitch phase consistency resilient to codecs.

• Court Fraud Disputes --> Strategy: ■ Extra Blockchain Voice Pass stamps SHA-256 Merkle root on Polygon.

• Sore Throat / Cold False Positives --> Strategy: ■ Extra Behavioral Biometrics scoring with Step-Up OTP.



IMPACT AND BENEFITS — Social, Economic & Legal

- • Social Impact: Restores public trust in phone calls; VIP Shield prevents IAS/IPS officer extortion.
- • Economic Impact: Prevents multi-crore wire transfer fraud in banking call centers.
- • Legal Impact: Blockchain Voice Pass provides tamper-proof court evidence; 100% DPDP Act & RBI compliant.



RESEARCH AND REFERENCES

- **ASVspoof 2021/2023 Benchmarks:** Official synthetic speech evaluation datasets.
- **India DPDP Act 2023:** Rules for Zero Raw Audio retention & privacy compliance.
- ■ **RBI Cyber Security 2023:** Bank guidelines for multi-factor voice authentication.
- **PyTorch ECAPA-TDNN & SHAP:** Research papers on speaker embeddings & Explainable AI.
- **Polygon Web3 Smart Contracts:** Blockchain standards for SHA-256 Merkle court proof.

